



ADVANCED CLASSICAL DYNAMICS
MASTER 2 COMPU**PHYS**

Hamiltonian Chaos: Numerical Analysis of the Chirikov Standard Map

Ali FADLALLAH

Contents

1	Introduction	2
2	System Description	2
3	Twist map ($K=0$)	2
4	Chirikov Standard Map ($K>0$)	2
5	Phase Portrait Analysis	4
5.1	Phase Portraits for Successive Values of K	4
5.2	Resonances and the Onset of Chaos	11
5.3	Lyapunov Exponents	11
6	Conclusion	12

1 Introduction

In this report, we explore the dynamics of the Chirikov Standard Map, a paradigmatic example of a Hamiltonian system exhibiting chaotic behavior. The map is defined by a pair of iterative equations involving two variables, I and θ , and a parameter K . Our aim is to numerically investigate the behavior of this system for different values of K , focusing on the emergence of chaos and the structure of the phase space.

2 System Description

The Chirikov Standard Map is defined by the following equations:

$$\begin{cases} I_{n+1} = I_n + K \sin \theta_n & \text{mod}(2\pi) \\ \theta_{n+1} = \theta_n + I_{n+1} & \text{mod}(2\pi) \end{cases} \quad (2.1)$$

Where $I \in [0, 2\pi[$, $\theta \in [0, 2\pi[$, are variables representing action and angle respectively, and $K \in \mathbb{R}^+$ is a parameter controlling the strength of the nonlinearity in the system.

3 Twist map (K=0)

Sure, let's consider the Chirikov Standard Map (SM) for $K = 0$. In this case, the map simplifies to a twist map:

$$\begin{cases} I_{n+1} = I_n \equiv I_0 & \text{mod}(2\pi) \\ \theta_{n+1} = \theta_n + I_0 & \text{mod}(2\pi) \end{cases}$$

For a p -periodic point, the system returns to its initial state after p iterations. This means that

$$\begin{cases} I_{n+p} = I_n \\ \theta_{n+p} = \theta_n \end{cases}$$

Given that $I_{n+1} = I_n \equiv I_0$, we can see that I is constant for all n . Therefore, for any p -periodic orbit, I will be the same for all points on the orbit.

As for θ , we have $\theta_{n+1} = \theta_n + I_0$. After p iterations, we get $\theta_{n+p} = \theta_n + pI_0$. For the orbit to be p -periodic, we need $\theta_{n+p} = \theta_n$, which implies $pI_0 = 2\pi m$ for some integer m .

Therefore, the value of I in the $[0, 2\pi[$ domain for a p -periodic orbit is given by:

$$I = \frac{2\pi}{p} \quad \text{for } p \in \mathbb{Z}$$

4 Chirikov Standard Map (K>0)

Now, let's consider the Chirikov Standard Map (SM) for $K > 0$:

$$\begin{cases} I_{n+1} = I_n + K \sin \theta_n & \text{mod}(2\pi) \\ \theta_{n+1} = \theta_n + I_{n+1} & \text{mod}(2\pi) \end{cases}$$

The fixed points, of the map are the points (I^*, θ^*) that satisfy:

$$\begin{cases} I_{n+1} = I_n = I^* \\ \theta_{n+1} = \theta_n = \theta^* \end{cases}$$

Substituting these into the equations gives:

$$\begin{cases} I^* = I^* + K \sin \theta^* \\ \theta^* = \theta^* + I^* \end{cases} \quad (4.1)$$

Solving these equations gives the fixed points.

$$\begin{cases} K \sin \theta^* = 0 \\ I^* = 0 \end{cases}$$

$\sin \theta^* = 0$, has solutions:

$$\theta^* = 0 \quad \text{or} \quad \theta^* = \pi$$

Thus, the fixed points are:

$$(I^*, \theta^*) = (0, 0) \quad \text{and} \quad (0, \pi)$$

Stability of the Fixed points

To study the stability of these fixed points, we need to look at the Jacobian matrix of the map at the fixed points and compute its eigenvalues. If the absolute values of all eigenvalues are less than 1, the fixed point is stable; if at least one eigenvalue has an absolute value greater than 1, the fixed point is unstable.

The Jacobian matrix of the map is:

$$J(I, \theta) = \begin{bmatrix} \frac{\partial I_{n+1}}{\partial I_n} & \frac{\partial I_{n+1}}{\partial \theta_n} \\ \frac{\partial \theta_{n+1}}{\partial I_n} & \frac{\partial \theta_{n+1}}{\partial \theta_n} \end{bmatrix} = \begin{bmatrix} 1 & K \cos \theta_n \\ 1 & 1 - K \cos \theta_n \end{bmatrix} \quad (4.2)$$

At the fixed points, this becomes:

$$\text{At } (0, 0): J_{(0,0)} = \begin{bmatrix} 1 & K \\ 1 & 1 - K \end{bmatrix}$$

$$\text{At } (0, \pi): J_{(0,\pi)} = \begin{bmatrix} 1 & -K \\ 1 & 1 + K \end{bmatrix}$$

The stability of a fixed point depends on the eigenvalues of the Jacobian matrix at that point. In general, if any eigenvalue has a positive real part, the fixed point is unstable (hyperbolic).

The eigenvalues of this matrix are the roots of the characteristic polynomial

$$P(\lambda) = \lambda^2 - \text{tr}(J)\lambda + \det(J) \quad (4.3)$$

the (0,0) fixed point:

$$\lambda^2 - (2 - K)\lambda + 1 - 2K = 0$$

solving,

$$\lambda_{\pm}^{(0,0)} = 1 - \frac{K}{2} \pm \sqrt{K + \frac{K^2}{4}}$$

$K + \frac{K^2}{4}$ is always positif for $K > 0$, thus $\lambda_{\pm}$ is always real.

λ_+ is always positive (> 1) for $K > 0$. While

$$\begin{aligned} 0 \leq \lambda_- < 1 & \text{ for } 0 < K \leq 0.5 \\ \lambda_- < 0 & \text{ for } K > 0.5 \end{aligned}$$

Thus, in the region of $0 < K \leq 0.5$, both eigenvalues are real and positive, the **(0,0)** fixed point is an **unstable Node**.

for $K > 0.5$, both eigenvalues are real and of opposite sign, **(0,0)** fixed point is an **unstable Saddle**.

the (0, π) fixed point:

$$\lambda^2 - (2 + K)\lambda + 1 + 2K = 0$$

solving,

$$\lambda_{\pm}^{(0,\pi)} = 1 + \frac{K}{2} \pm \sqrt{K - \frac{K^2}{4}}$$

$\lambda_{\pm}$ is always positive for $0 < K \leq 4$. for $K > 4$ the $K - \frac{K^2}{4}$ term becomes negative and both eigenvalues becomes complex numbers with real positive parts.

Thus, for $0 < K < 4$, the **(0, π)** fixed point is an **unstable Node**.

For $K > 4$, the **(0, π)** fixed point is an **unstable Focus (Spiral) Point**.

Stability:

(0,0) is an unstable Node for $0 < K \leq 0.5$, and an unstable saddle for $K > 0.5$.

(0, π) is an unstable Node for $0 < K \leq 4$, and an unstable focus for $K > 4$

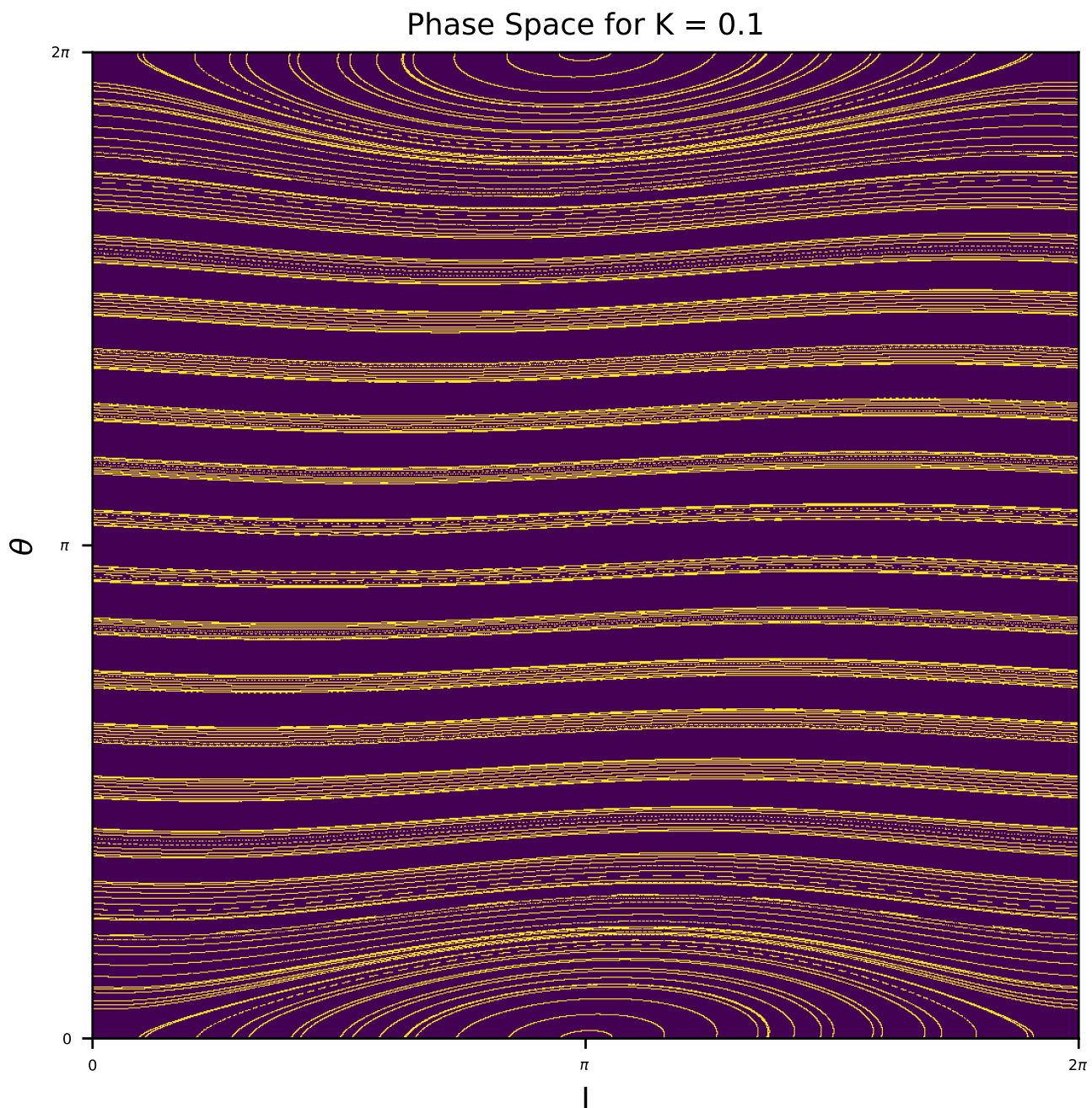
5 Phase Portrait Analysis

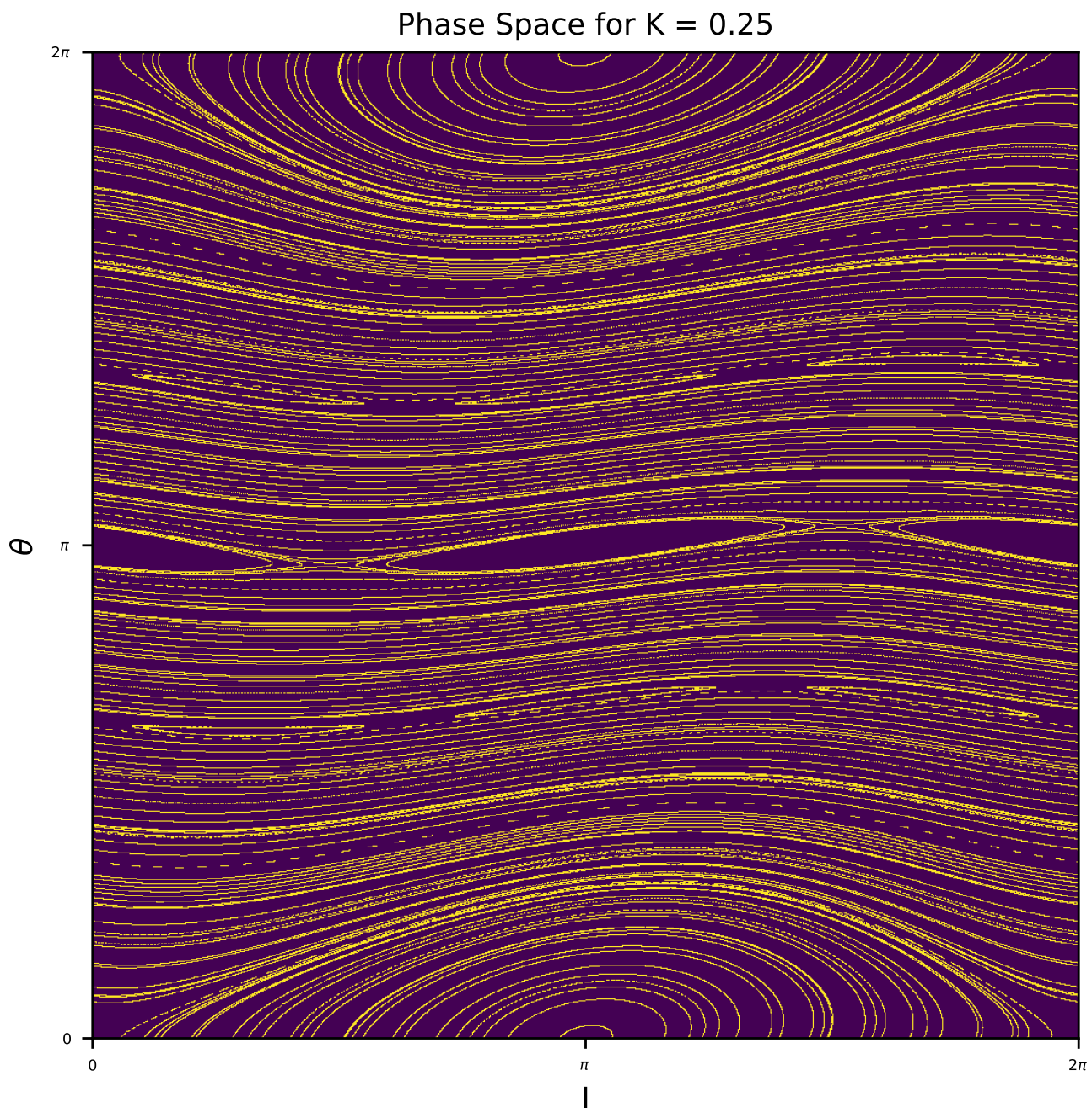
In this section, we numerically generate and analyze the phase portraits of the Chirikov Standard Map for various values of the parameter K .

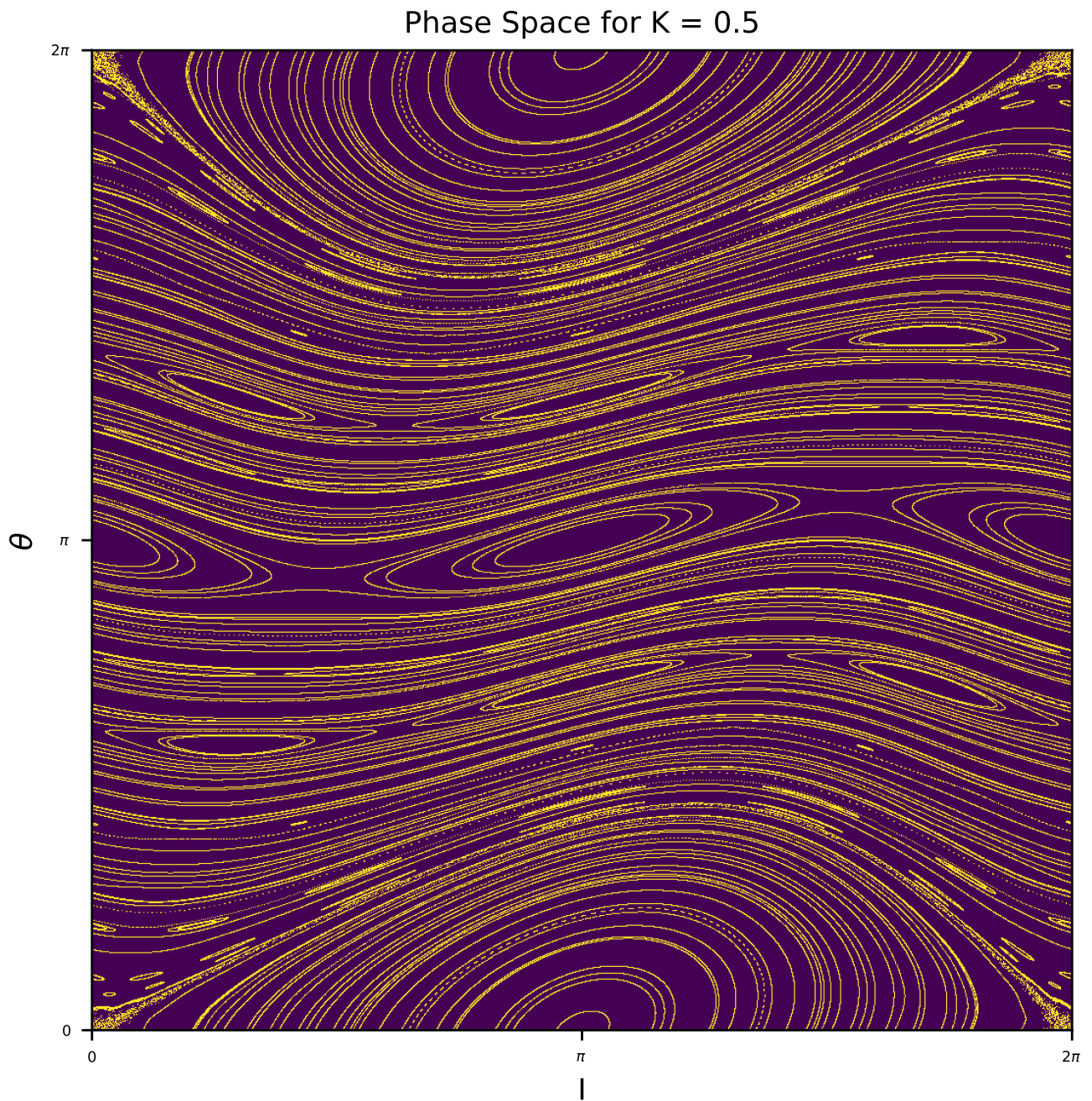
5.1 Phase Portraits for Successive Values of K

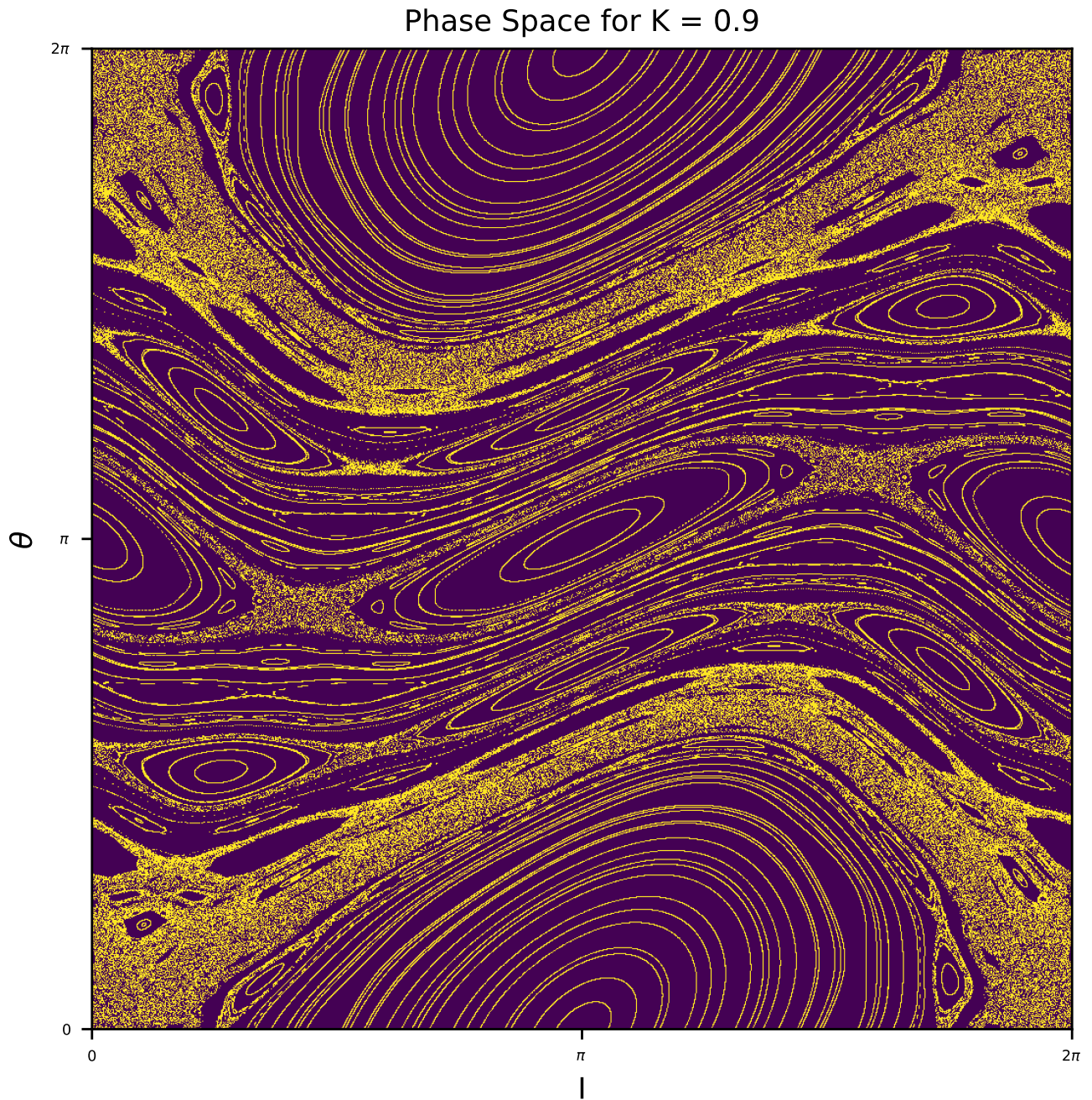
We start by generating phase portraits for successive values of K . For each value of K , we construct an initial grid spanning the $[0, 2\pi[\times [0, 2\pi[$ domain and iterate the Chirikov Standard Map a large number of times. The resulting set of points gives us an approximation of the phase portrait of the map.

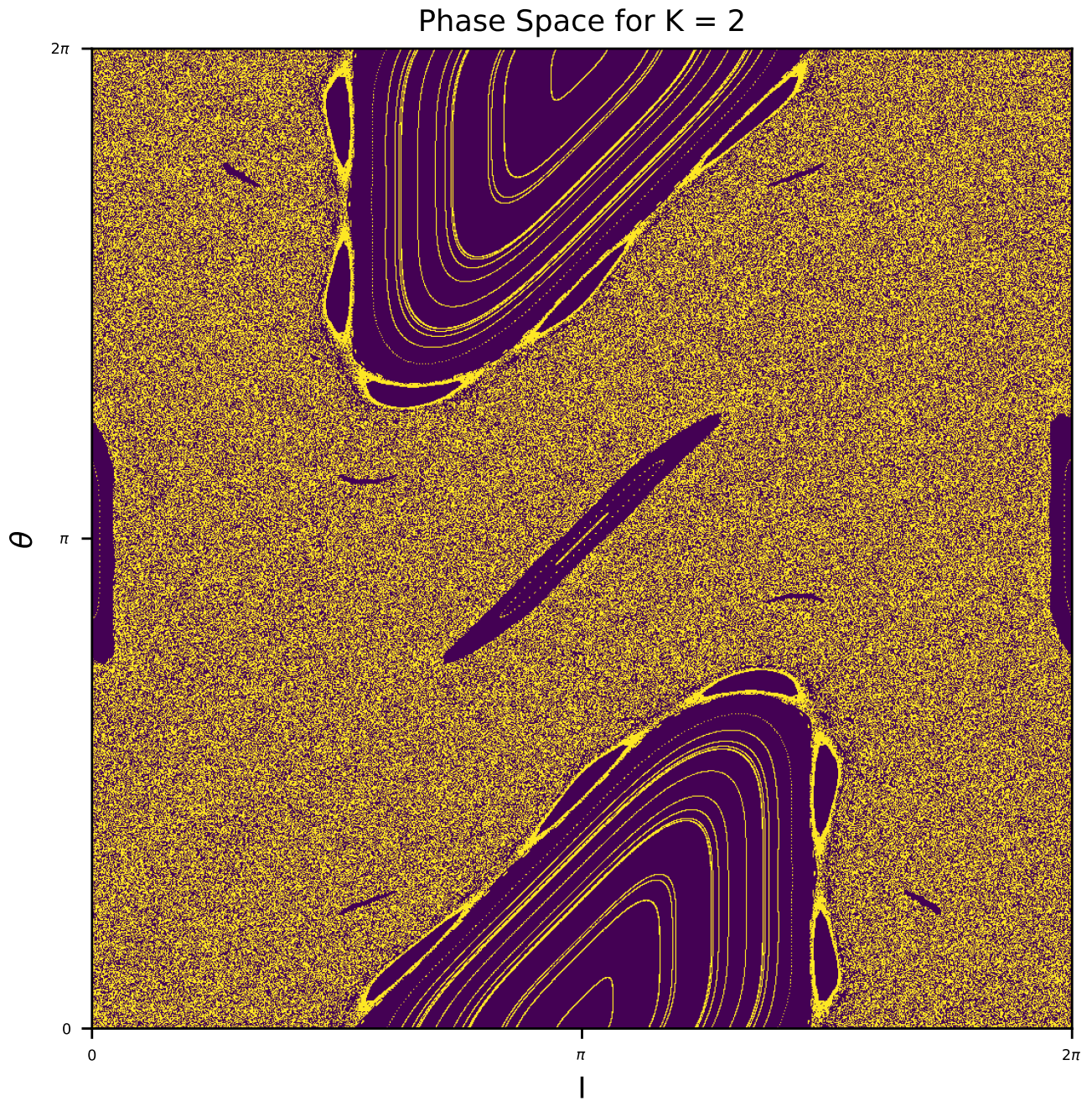
The following figures were generated using an initial grid of size $M=20$, iterating the map $N=2000$ times, and using an array of size $L=1000$ to represent the phase space.











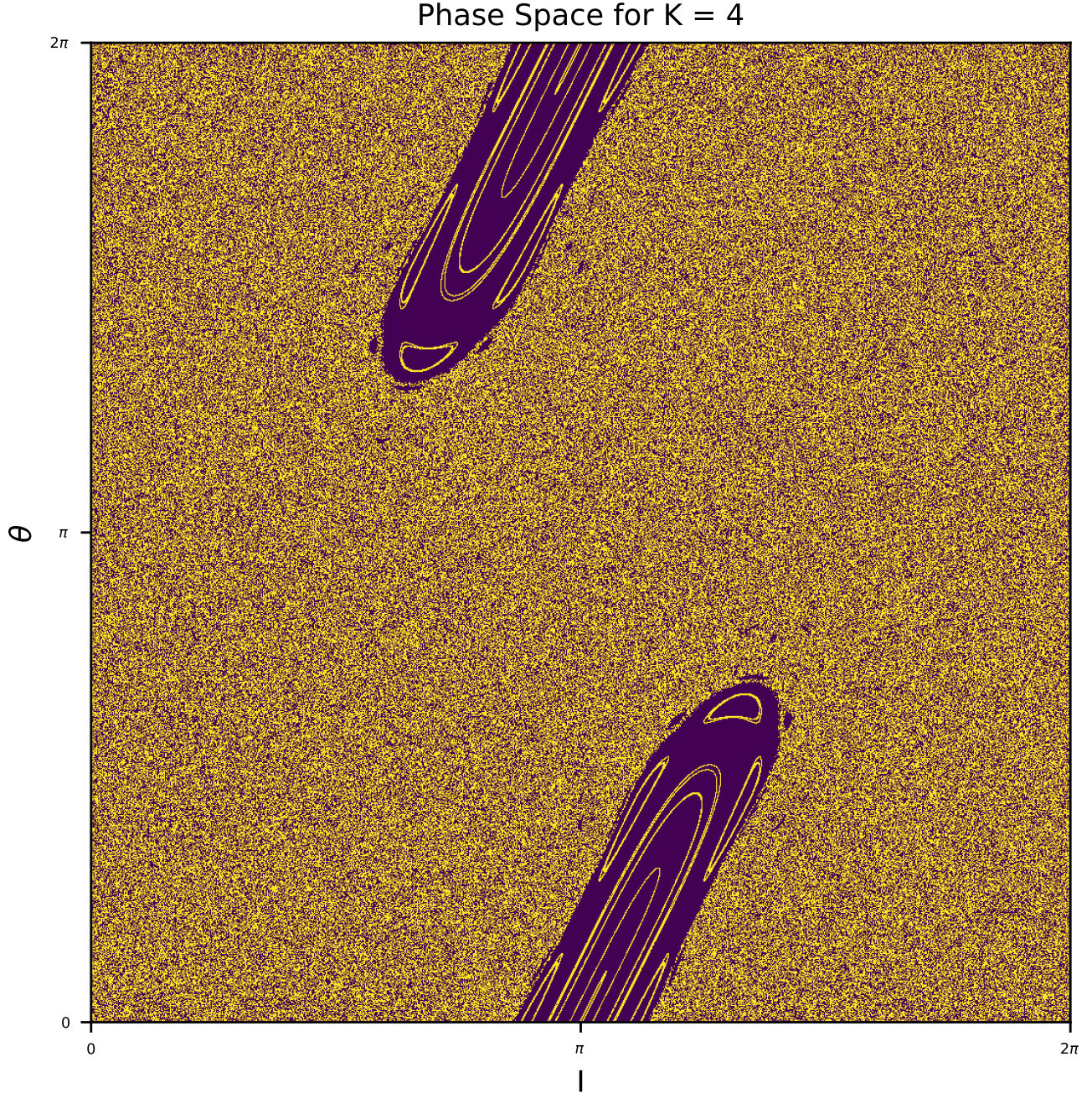


Figure 1: Phase Portrait of the Chirikov Standard Map for $K > 0$. Each point in the phase space represents a unique state of the system. The color of each cell represents the number of times an orbit visits that region of the phase space. The figures were generated using an initial grid of size $M = 20$, iterating the map $N = 2000$ times, and using an array of size $L = 1000$ to represent the phase space.

Figure 1 shows the phase portrait of the Chirikov Standard Map for $K = 0.1, 0.25, 0.5, 0.9, 2, 4$.

As K increases, we observe the emergence of intricate structures in the phase space, indicative of the onset of chaos. These Figures shows the phase space of the Chirikov Standard Map for a positive value of K . The emergence of

intricate structures in the phase space is indicative of the onset of chaos. The color of each cell is determined by the mean Lyapunov exponent of the orbits that visit that cell, providing a measure of the rate of divergence of nearby orbits.

As seen in the generated figures, for small values of K , the phase portrait consists mainly of regular, quasi-periodic orbits. As K increases, more and more resonances become affected, leading to the appearance of chaotic regions in the phase space. For even larger values of K , the chaotic regions begin to dominate the phase space, indicating the onset of global chaos.

Self-Similarity of the Phase Portrait: Upon close inspection, we find that the phase portrait of the Chirikov Standard Map exhibits a property of self-similarity. As we zoom into the phase space, we observe that the structure of the phase space at a smaller scale resembles the structure at the larger scale.

5.2 Resonances and the Onset of Chaos

We observe that:

- **for weak K , e.g., $K = 0.1$,** the first affected resonance is the $1 : 1$ resonance. the map is close to being integrable and the dynamics are nearly regular.
- **If K increases a little more,** higher-order resonances mainly $2 : 1$, become affected.
- **For K values around 0.5 ,** several other resonances become visible up to $5 : 1$.
- **Around $K=4$:** At $K=4$, the map undergoes a bifurcation and the phase space becomes fully chaotic. This means that almost all initial conditions will lead to chaotic trajectories.
- **K_c ,** through observations of the maps at different value of K , we estimate that the main chaotic component spans the whole domain $I \in [0, 2\pi)$ for $K \approx 0.9$. This is the critical value K_c at which the transition to global chaos occurs. For K around this value, the phase portrait shows a mix of regular islands and chaotic seas. As K approaches K_c from below, the chaotic component gradually expands to fill the entire I domain.

5.3 Lyapunov Exponents

Here, we study the Chirikov Standard Map for $K = 9$ by analyzing the Lyapunov exponents in the phase space. The Lyapunov exponents provide a measure of the rate of divergence of nearby orbits, and hence, a measure of the chaos in the system.

Figure 2 shows the Lyapunov exponents of the Chirikov Standard Map for $K = 9$. Regions with positive Lyapunov exponents, represented by warmer colors (red-ish), indicate chaotic behavior, while regions with negative or zero Lyapunov exponents, represented by cooler colors (blue-ish), indicate regular or periodic behavior. we observe that the chaotic regions are more prevalent, indicating the onset of widespread chaos in the system.

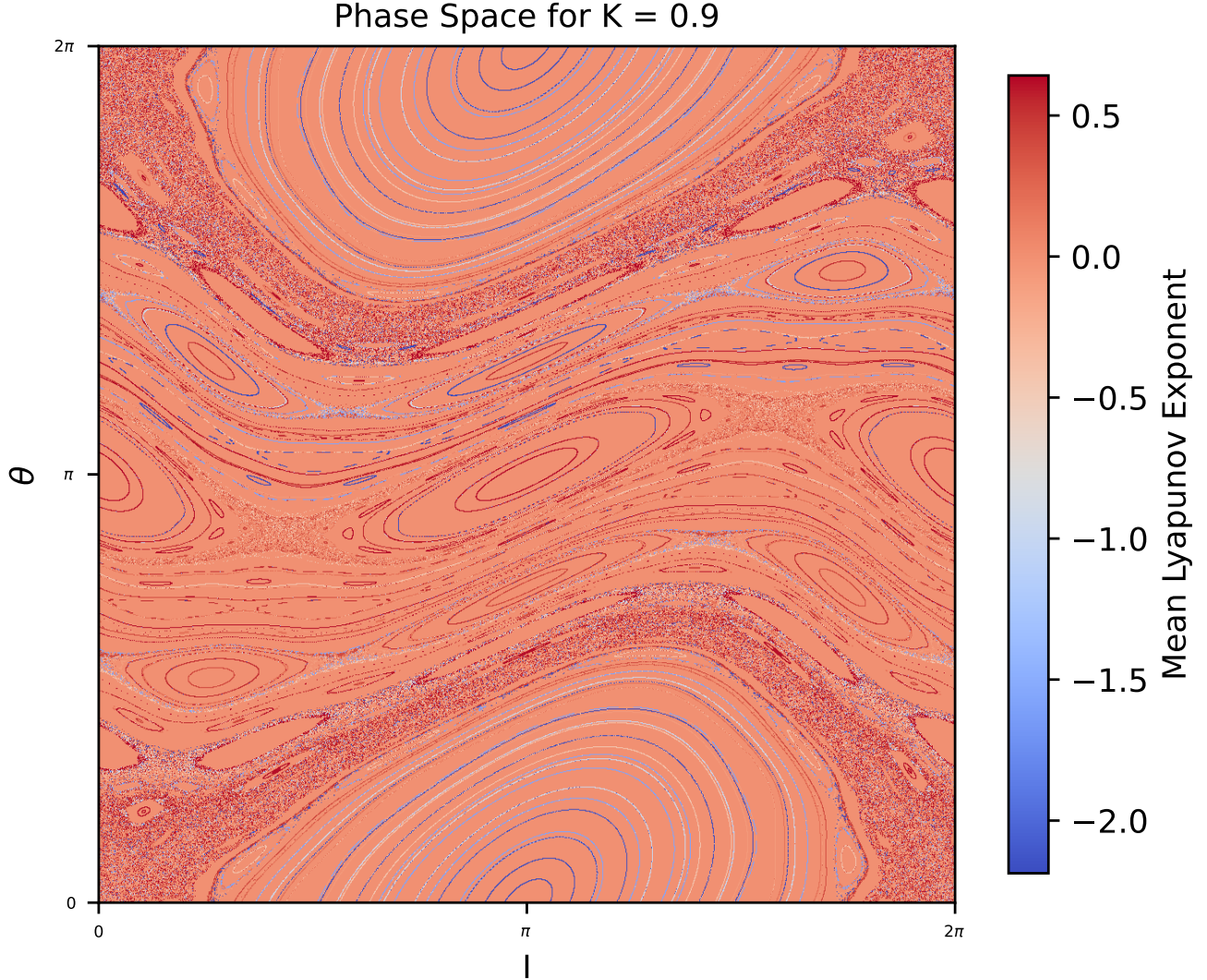


Figure 2: Lyapunov Exponents of the Chirikov Standard Map for $K = 0.9$. The color of each cell represents the mean Lyapunov exponent of the orbits that start from that region of the phase space. The figure was generated using an initial grid of size $M = 20$, iterating the map $N = 2000$ times, and using an array of size $L = 1000$ to represent the phase space.

6 Conclusion

Through our numerical investigations, we observed the onset of chaos in the Chirikov Standard Map as the parameter K was varied. For small K , the system exhibited regular, quasi-periodic behavior, with trajectories in phase space forming closed curves. As K was increased, these curves began to break up, signaling the onset of chaos. For sufficiently large K , the phase space was dominated by a chaotic sea, indicating fully developed chaos. We also inspected the degree of chaos through the Lyapunov exponents.

These observations highlight the rich dynamics of the Chirikov Standard Map.